



A Review on Current Status and Methodology of UAV Swarm Combat Effectiveness Evaluation

Wenjia Wang, Jialong Zhang, Hanzhang Xue, Li Yu, and Lei Ye^(✉)

Test Center, National University of Defense Technology, Xi'an 710102, China
yelei@nudt.edu.cn

Abstract. This paper systematically summarizes the research development in the field of UAV swarm combat effectiveness evaluation and integrates the development of the evaluation index system and methodology through a multidimensional categorization framework. First, in terms of the evaluation index system, the existing studies are categorized into a three-dimensional evaluation framework oriented to capability factors, environment factors, and combat missions. Second, at the technical level of effectiveness evaluation methodology, existing studies are divided into four categories: qualitative analysis based on expert experience, modeling methodology based on mathematical analysis, simulation-based dynamic deduction technology, and data-driven and artificial intelligence methodology. Finally, this paper analyzes the shortcomings of the current research in both evaluation system robustness and dynamic adaptive evaluation, and proposes that the future trends are to develop the multi-mission collaborative evaluation index system, the dynamic evaluation model, the intelligence-driven evaluation-optimization closed-loop mechanism, and other related research. The results can provide theoretical support for UAV swarm combat effectiveness evaluation research.

Keywords: UAV Swarm · Effectiveness Evaluation · Index System · Evaluation Methodology

1 Introduction

In recent years, the breakthroughs in artificial intelligence and group intelligence algorithms have pushed the UAV to evolve towards clustering and autonomy, which have jointly produced the new combat mode of UAV swarm combat [1]. The core concept of UAV swarm combat refers to a combat system formed by multiple UAVs through autonomous coordination and intelligent networking, with the support of a networked information system to achieve a centralized and unified combat capability [2]. UAVs within a swarm can fly autonomously to the mission area while avoiding collisions with other units, and can automatically detect threats and locate targets based on artificial intelligence and image processing technology.

With the continuous development of the intelligence and control technology of UAV swarm, it has gradually formed a combat mode based on the theory of OODA (observation, orientation, decision, action) [3]. In order to measure the UAV swarm in the

combat environment, it is necessary to analyze the combat capability, combat suitability, technical performance, and mission completion rate to realize the reliable evaluation of UAV swarm combat effectiveness. The UAV swarm combat effectiveness evaluation based on high-confidence evaluation indexes and models, which can obtain the effective description of UAV swarm autonomous and cooperative characteristics, and provide a basis for whether UAV swarm meet the requirements.

The core steps of the UAV swarm combat effectiveness evaluation mainly include the following points. First, the purpose and scope of the evaluation are clarified in the light of the combat mission. Second, starting from the top-level index (i.e., combat effectiveness), determine the overall framework of the evaluation, and construct a combat effectiveness evaluation index system. Third, design the evaluation method and establish the evaluation model. Finally, calculate and analyze the UAV swarm combat effectiveness evaluation and conduct experimental verification and analysis. UAV swarm combat effectiveness evaluation is characterized by emergence, dynamics, and a nonlinear coordination mechanism when performing combat missions, which makes the evaluation difficult in the design of evaluation indexes and high in the requirements of evaluation models.

2 Research on UAV Swarm Combat Effectiveness Evaluation Index System

2.1 The Capability-Based Evaluation Index System

The evaluation index system is the basic framework of UAV swarm combat effectiveness evaluation, and its scientificity and completeness directly affect the reliability of the evaluation result. The capability-based UAV swarm combat effectiveness evaluation index system, i.e., the basic parameter indexes that enable the swarm to operate safely and stably, and complete the mission successfully. This type of evaluation index system includes the UAV's survivability, maneuverability, perception, communication, and combat capability, as well as autonomous decision-making capability, cooperative interaction capability, and continuous learning capability, etc., which are inherent, intrinsic, and controllable elements of the swarm. A number of scholars have already researched this problem [4–9].

It can be concluded from the analysis of existing studies that the combat capability of UAV swarm is based on their individual performance and the overall performance of the swarm, and current research has formed a consensus on the core capability index, focusing on the dimensions of survivability, cooperative interaction, and autonomous decision-making, and has verified the basic efficacy through simulation and small-scale tests. However, most of the existing researches consider the inherent attributes of UAV swarm, focusing on the construction of UAV swarm equipment performance index system, and less consideration is given to the degree of influence of the UAV swarm under the complex combat environment or the effect of the capability in the complex confrontation environment.

2.2 The Environment-Based Evaluation Index System

Environmental factors are externally imposed conditions and constraints, such as terrain, weather, electromagnetic interference, and other uncontrollable factors. UAV swarm will be affected by the surrounding environmental factors when it carries out the mission, and the external conditions in which the swarm is located have a direct impact on the UAV swarm combat effectiveness. Authors in [10–14] have conducted in-depth studies on such problems.

By analyzing the existing research, it can be concluded that the current research is fruitful at the level of environment modeling and intelligent algorithms. The construction of the evaluation index system oriented to environmental factors takes into account the environmental adaptability and appropriateness of use of the UAV swarm, which can measure the combat capability of the UAV swarm in a more specific way. However, the existing research in dynamic confrontation adaptability, large-scale swarm robustness, and other aspects of the research is not perfect; the construction of the evaluation index system should be combined with combat simulation and practical exercises.

2.3 The Combat Mission-Based Evaluation Index System

Combat missions are the demands of the application scenarios, involving specific combat objectives, mission complexity, threat level, and other variables. The construction of the combat mission-based evaluation index system can reflect the utilization of UAV swarm combat capability in various mission segments and combat phases in the context of combat missions. This type of problem has been studied in the literature [15–20].

By summarizing the relevant research, it can be seen that the evaluation index system for combat missions has made significant research achievements in terms of hierarchical structure and specialized index. The relevant research mainly focuses on evaluating the performance of UAV swarm with regard to the mission requirements. However, when analyzing the combat effectiveness, the reliability of UAV swarm in different combat environments and the cooperation with combat commanders should also be taken into account to build a more perfect evaluation index system.

3 Development of UAV Swarm Combat Effectiveness Evaluation Methodology

3.1 Experience-Based Evaluation Methodology

Experience-based evaluation methodology mainly refers to methods based on human subjective experience or specialized knowledge. The UAV swarm combat effectiveness evaluation is conducted through the subjective experience of domain experts to judge the impact of each evaluation index, combined with qualitative analysis and quantitative calculation.

- (1) The analytic hierarchy process (AHP) method is to decompose complex problems into a hierarchical structure, through subjective experience to determine the weight of the indexes, and calculate the total score. In UAV swarm combat effectiveness evaluation, AHP can effectively integrate various types of indexes and establish a hierarchical evaluation system to provide evaluation results [21–23].

- (2) The fuzzy comprehensive evaluation method uses the fuzzy relationship synthesis theory to deal with uncertainty and ambiguity in evaluation. By synthesizing the fuzzy evaluation values and weights of the effectiveness evaluation indexes, the qualitative evaluation is transformed into the quantitative evaluation [24–26].

The advantage of the experience-based evaluation method is that it can effectively integrate quantitative and qualitative indexes, structurally deal with the complex relationships between indexes, be simple to calculate, and reflect the preferences of decision makers. However, the determination of weights is highly subjective, and it is difficult to ensure that each index can be quantified through experimentation.

3.2 Mathematical-Based Evaluation Methodology

The mathematical modeling-based evaluation method is to take the mathematical model as the basis, according to the meaning of the evaluation indexes and the relationship between them, analyze the internal connection of each index, and establish the mathematical evaluation model.

- (1) The ADC method uses three elements, A(availability), D(dependability) and C(capability), to evaluate the comprehensive indexes: $E = ADC$. The ADC method analyzes the problem logically, and some scholars have evaluated the UAV swarm combat effectiveness using this method [27, 28].
- (2) The system effectiveness analysis method defines the swarm capability and mission requirement in the combat environment, respectively, and calculates effectiveness by comparing the intersection set of the two. This method focuses on the matching degree between the capability and requirement, and is suitable for evaluating whether or not a UAV swarm can satisfy a particular mission [29, 30].
- (3) The data envelopment analysis method is a linear programming method for evaluating the relative effectiveness of units in a complex system. This method can be used to compare the combat effectiveness of different UAV swarms and different combat plans, and provide specific directions for improvement [31, 32].

The advantages of mathematical modeling-based evaluation methods include efficient calculation, interpretable results, and the ability to compare multiple combat plans. However, such a method is limited to modeling complex systems, and it is difficult to comprehensively show the performance of the UAV swarm.

3.3 Simulation-Based Evaluation Methodology

The simulation-based evaluation method is to build a simulation model that covers the UAV swarm, combat environment, and mission process according to the combat mission, and then run a large number of simulation experiments and output the statistical results to evaluate the combat effectiveness.

- (1) The Monte Carlo method simulates the random phenomena that may occur in real combat through multiple random sampling, and then summarizes the simulation experimental data obtained, analyzes the distributions of mean, variance, and confidence intervals to obtain the evaluation result of UAV swarm [33, 34].

- (2) The system dynamics method describes the interactions of elements through causal analysis. In UAV swarm combat effectiveness evaluation, this method effectively solves the complexity and uncertainty problems by quantifying the dynamic coupling relationship among capability, environment, and mission [19, 35].
- (3) The simulation verification method based on the architecture framework mainly adopts DoDAF/UAF and other frameworks to describe the structure, behavior, and interaction of the combat system. Through simulation and data analysis, it can effectively solve the problem of UAV swarm combat effectiveness optimization [36, 37].

The advantage of the simulation-based evaluation method is that it can deal with highly complex, dynamic, and nonlinear systems by establishing a simulation model that is adapted to actual combat mission and combat environment. The shortcoming is that the development of the model requires a large amount of data and a high computational overhead.

3.4 Data-Driven and Artificial Intelligence-Based Evaluation Methodology

Data-driven and artificial intelligence evaluation methods derived from big data and artificial intelligence technology refer to the application of a large amount of data to train models and establish relationships from input to output, which applies to performance prediction and effectiveness evaluation.

- (1) The neural network method is a machine learning algorithm that is trained to adaptively deal with complex nonlinear relationships. When using this method to evaluate UAV swarm combat effectiveness, first, train the optimal neural network evaluation model through a large amount of data, and then output the evaluation result by taking the obtained index values as inputs [38, 39].
- (2) The support vector machine method is to transfer problems from the input space to a high-dimensional space through a kernel function nonlinear mapping. This method can accurately describe the interactions between UAV swarm combat effectiveness indexes and is suitable for solving prediction problems [40, 41].
- (3) The Bayesian network method is an uncertainty inference method based on a probabilistic graphical model. The UAV swarm combat effectiveness evaluation can use directed acyclic graphs to structurally characterize the conditional dependencies between elements within the swarm and quantify the uncertainty transfer mechanism based on conditional probability tables. In addition, this method can analyze the influence of each index on the overall effectiveness through the inference [16, 42].

Data-driven and artificial intelligence evaluation methods have strong prediction and evaluation capability, which are of great value to the UAV swarm combat effectiveness evaluation. Noting that this type of method requires a large amount of high-quality training data to ensure the accuracy and reliability of the evaluation results.

4 Existing Research Gaps and Future Research Perspectives

4.1 Gaps in Existing Research

Although UAV swarm combat effectiveness evaluation has made great progress, there are still significant limitations in terms of the improvement of the index system, the innovation of the modeling technology, and the interpretability of the evaluation results, which constrain the application in real-world combat decision-making.

- (1) The effectiveness evaluation lacks standardization and robustness. The current evaluation has not yet formed a unified structure covering “capability-environment-mission”, and the construction of the existing index system is mostly based on traditional combat scenarios, making it difficult to support the new combat missions. In addition, the existing evaluation mostly relies on ideal data sources, and it is difficult to deal with the problem of key indexes being unavailable due to interference in a combat environment. A scientifically valid mechanism for alternative indexes is lacking, and the robustness of the evaluation is insufficient. The interpretability of the results of existing effectiveness evaluation also needs to be improved.
- (2) The existing UAV swarm combat effectiveness evaluation has limitations in practical applications. In terms of dynamic adaptability, the traditional static model is difficult to portray the time-varying combat environment and swarm state evolution. In terms of solving efficiency, most of the methods find it difficult to balance the accuracy and solving scale, and solution efficiency tends to decay exponentially with increasing size. In terms of uncertainty quantification, probabilistic methods are overly dependent on a priori assumptions and ignore cognitive uncertainty. In terms of method synergy, various evaluation methods can only achieve better results in specific scenarios, and hybrid evaluation methods are less frequently applied.

4.2 Prospects for Future Research

In view of the current study of UAV swarm combat effectiveness evaluation and the deficiencies that exist, combined with the future combat missions and the development of intelligent technology, this problem can be researched in the future from the following aspects.

- (1) Establish a multi-mission coordinated evaluation index system. In the future, UAV swarm needs to complete many different combat missions, therefore, it is necessary to consider diversified mission requirements when constructing the UAV swarm combat effectiveness evaluation index system, realize the expansion from single-mission to multi-mission cooperation, and establish an expandable effectiveness evaluation index system that applies to many different combat missions.
- (2) Dynamic effectiveness evaluation research. The existing UAV swarm combat effectiveness evaluation is mainly based on static evaluation, in the future, when constructing the effectiveness evaluation, it should focus on the real-time and dynamic characteristics of UAV swarm under the real-word combat environment, and combine the real-time missions, combat objects, etc., to establish a dynamic model and realize the dynamic evaluation of UAV swarm combat effectiveness.

- (3) Intelligence-driven evaluation-optimization closed-loop research. By taking the results of the effectiveness evaluation as the feedback information, an evaluation-optimization closed-loop feedback mechanism can be constructed. Combined with intelligent algorithms, the iteration of real-time effectiveness evaluation and dynamic decision-making will be carried out to achieve dynamic optimization of effectiveness, promote the UAV swarm combat in the direction of autonomous evolution, and provide technical support for future unmanned combat.

5 Conclusion

This paper focuses on the core issues of UAV swarm combat effectiveness evaluation, through systematically analyzing the construction of the evaluation index system and the technical development of the evaluation method, reviewing and classifying the existing research, and analyzing the gaps of current research and the development trend in the future. The research results can provide technical reference for relevant personnel to carry out the unmanned combat effectiveness evaluation.

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